

RESTORATION ESSENTIALS

Master Restoration Checklist

The bumper-to-bumper checklist for 1953–1973
American muscle & classic cars

12	phases, teardown to final inspection
101	hands-on checkpoints with sign-off boxes
1	binder-ready system: print it, check it, file it

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How to use this checklist

Print it, put it in a binder, and work it phase by phase. Each checkpoint has a box — check it off only when the work is actually done and verified, not when it's 'close enough.'

The checklist follows the order a real restoration runs: assess, plan, tear down, body, paint, chassis, drivetrain, electrical, interior, reassembly, first start, final inspection. Some phases overlap in a busy shop; the order still keeps you from painting before the metalwork is proven.

Write dates and initials next to completed phases. When the car is done, this checked-off document goes in the binder with your receipts and photos — it's part of the car's provenance and value.

One rule covers everything: **when in doubt, stop and verify.** The most expensive mistakes in restoration are the ones discovered two phases too late.

PHASE 1 OF 12

ASSESSMENT & DOCUMENTATION

Do this before a single wrench turns. Everything downstream depends on what you record now.

- ☐ Photograph the car from all four corners, roof, underhood, trunk, interior, and undercarriage — minimum 100 photos, dated.
- ☐ Record the VIN, cowl/fender tag data, engine pad stamp, transmission tag, and axle code. Photograph every stamping in place.
- ☐ Decode all tags and establish numbers-matching status **before** disassembly.
- ☐ Photograph every factory sticker, inspection mark, chalk mark, and crayon notation — these are restoration gold.
- ☐ Grade overall condition on a 1–6 scale and write it down. This sets the target: driver, show, or concours.
- ☐ Check for prior accident damage: uneven panel gaps, overspray lines, welded seams, misaligned frame horns.
- ☐ Map every rust area on a body diagram. Probe with an awl — distinguish surface rust from perforation.
- ☐ Inspect the five rust magnets: floors, trunk floor, rockers, lower quarters, cowl and windshield channels.
- ☐ Start a build binder now: one section per system, plastic sleeves for every receipt.
- ☐ Take a compression test and record the numbers — baseline engine health before teardown.
- ☐ Confirm the title is clear and transferable into your name **before** spending a dollar on parts.
- ☐ Note the car's build date versus part date codes — a two-month spread is normal; two years is a red flag.

Shop tip: Bill's rule of thumb: the camera is the cheapest tool in the shop. You will never regret taking too many photos.

PHASE 2 OF 12

PLANNING & BUDGET

A restoration without a plan is just expensive disassembly.

- ☐ Define the target in writing: daily driver, weekend show car, or concours. Every decision flows from this.
- ☐ Research correct finishes, fasteners, and part numbers for your exact year and model before buying anything.
- ☐ Price the big five first: body/paint, engine machine work, chrome plating, interior, tires and wheels.
- ☐ Add a 25–30% contingency on top. Rust always hides more rust.
- ☐ Decide DIY vs. outsource for each system — and book the good shops early; they all have wait lists.
- ☐ Build a parts-sourcing list with quality tiers: NOS, quality reproduction, and donor-car parts.
- ☐ Set a realistic timeline with milestones, not a single distant deadline.
- ☐ Arrange dry, secure storage for the body, parts, and fasteners for the full duration.

Shop tip: Buy the assembly manual and factory service manual for your car now if you don't own them — they pay for themselves in the first weekend.

PHASE 3 OF 12

TEARDOWN & CATALOGING

How the car comes apart determines how — and whether — it goes back together.

- ☐ Bag and tag EVERYTHING: zip bags, permanent marker, one bag per assembly. No exceptions.
- ☐ Photograph each assembly before removal and at each step of disassembly.
- ☐ Keep a fastener log: location, size, thread, finish — or use compartmented trays labeled by system.
- ☐ Drain all fluids before pulling the drivetrain: coolant, engine oil, transmission, differential, fuel.
- ☐ Support the body on rated jack stands or a rotisserie before removing suspension or drivetrain.
- ☐ Label every wire and hose at both ends with numbered tags; log the numbers.
- ☐ Store glass upright, padded, and labeled — it scratches if you look at it wrong.
- ☐ Separate parts into four streams: reuse, rebuild, replate/rechrome, replace.
- ☐ Keep all VIN-stamped and date-coded parts segregated — these never go in the generic bins.
- ☐ Verify jack stands are rated for the load; never work under a car supported by a jack alone.
- ☐ Cap fuel lines and plug every opening as you disconnect — dirt in an open line becomes a future failure.

Shop tip: *Your future self, six months from now, staring at a coffee can of bolts, will thank present-you for every label.*

PHASE 4 OF 12

BODY & METALWORK

The foundation of everything. Shortcuts here show up in the paint — and in the value.

- ☐ Strip to bare metal (chemical dip or media blast). You cannot fix what you cannot see.
- ☐ Re-map rust on bare metal. Cut out all perforation — never fiberglass or filler over rust.
- ☐ Butt-weld patch panels; grind welds flush. Body filler is a skim coat, not a structural material (1/8 in. max).
- ☐ Hang doors, hood, and decklid and set panel gaps BEFORE final bodywork — metal moves, filler doesn't.
- ☐ Test-fit all trim, glass, weatherstrip, and emblems before paint. If it doesn't fit now, it won't fit later.
- ☐ Epoxy-prime bare metal within hours of stripping — flash rust starts fast.
- ☐ Block-sand with guide coat until panels are laser-straight in raking light.
- ☐ Verify drip rails, cowl, and window channels are solid — leaks destroy interiors.

Shop tip: Gaps are set in metal, not in paint. Every hour spent fitting panels now saves three hours of regret later.

PHASE 5 OF 12

PAINT & FINISH

Paint is 10% spraying and 90% everything before it.

- ☐ Verify the paint code against the cowl tag. Spray test panels under your actual booth conditions.
- ☐ Follow product data sheets exactly: mix ratios, flash times, recoat windows.
- ☐ Paint jambs, underhood, trunk, and interior metal in their CORRECT finishes — not everything is body color.
- ☐ Document which products went on which panels for future repair matching.
- ☐ Allow full cure before cut-and-buff; then protect fresh paint during reassembly (fender covers, tape on edges).
- ☐ Reinstall stainless and chrome trim only after paint is fully cured.
- ☐ Hold booth temperature and humidity inside the product's specified range; log conditions for each session.

Shop tip: *The underside and jambs are where judges — and knowledgeable buyers — look first.*

PHASE 6 OF 12

CHASSIS, SUSPENSION, BRAKES & STEERING

Safety lives here. Nothing cosmetic matters if the car won't stop and steer.

- ☐ Inspect the frame for cracks, previous repairs, and straightness; measure diagonals.
- ☐ Rebuild or replace: ball joints, control-arm bushings, tie rods, idler arm, pitman arm.
- ☐ Brakes: new steel or NiCopp lines, hoses, wheel cylinders or calipers, master cylinder, booster check.
- ☐ Use factory-correct finishes: chassis black, natural steel, or plated — per your target level.
- ☐ Torque every suspension and brake fastener to specification with a calibrated torque wrench.
- ☐ Repack or replace wheel bearings; set preload correctly.
- ☐ Replace all rubber brake hoses regardless of appearance — age kills them from the inside.
- ☐ Check the steering box or rack for play; rebuild or replace — vague steering ruins the whole car.

Shop tip: *If it holds the car up, stops the car, or steers the car, it gets new or rebuilt. No judgment calls.*

PHASE 7 OF 12

ENGINE & DRIVETRAIN

Match the hardware to the stampings, and the build to the break-in.

- ☐ Photograph all stampings and casting numbers BEFORE machine work — decking can erase them.
- ☐ Machine-shop checklist: magnaflux, bore and hone with torque plate, deck, line-hone, crank polish or grind.
- ☐ Verify casting numbers and date codes fall in the correct window for your car's build date.
- ☐ Assemble with assembly lube; prime the oil system before the first crank.
- ☐ Flat-tappet cam break-in: 2,000+ rpm for 20 minutes, varying rpm, never idle — with high-zinc oil.
- ☐ Transmission: full rebuild or documented inspection; new seals, bushings, filter, and correct fluid.
- ☐ Rear axle: check gear pattern, new bearings and seals, correct ratio — keep the ratio tag.
- ☐ New motor and transmission mounts; inspect the driveshaft and U-joints.

Shop tip: *The two most expensive words in engine building are 'while we're in there' — said without a plan. With a plan, they're the cheapest.*

PHASE 8 OF 12

ELECTRICAL & WIRING*Fifty-year-old wiring is a fire waiting for a reason.*

- ☐ Replace or professionally rebuild the harnesses — engine, dash, forward lamp, and rear body.
- ☐ Test every circuit before the dash goes in for the final time.
- ☐ Route, clip, and loom wiring per the assembly manual — correctness and chafe protection in one.
- ☐ Grounds: clean to bare metal with star washers. Most 'electrical problems' are ground problems.
- ☐ Replace the voltage regulator, horn relay, and flasher as a matter of course.
- ☐ Verify charging output and battery draw (parasitic drain test) before calling it done.
- ☐ Replace the battery cables, not just the battery — corroded cables cause half of all no-starts.

Shop tip: Dielectric grease on every connection you touch. Future-you, troubleshooting at dusk, says thanks.

PHASE 9 OF 12

INTERIOR & TRIM*Fit everything dry first. Adhesive is a commitment.*

- ☐ Test-fit seat covers, door panels, and headliner before final installation.
- ☐ Rebuild and calibrate gauges; verify correct lenses, needles, and faces for your year.
- ☐ Fit all weatherstrip dry first, then install with the correct adhesive.
- ☐ Straighten, polish, or replating chrome and stainless trim — then test-fit before final install.
- ☐ Replace window felts, division-bar seals, and trunk weatherstrip while access is easy.
- ☐ Install carpet with proper padding; trim for shifter, seat tracks, and seatbelts before seats go in.
- ☐ Verify every interior light, switch, and accessory functions before trim panels close up access.
- ☐ Lubricate window regulators and lock mechanisms while access is open — you'll never have a better chance.

Shop tip: Interior work rewards patience more than any other phase. Rushing here is visible every time you open the door.

PHASE 10 OF 12

REASSEMBLY & FITMENT*This is where the bag-and-tag system pays for itself.*

- ☐ Reassemble in reverse order of teardown, system by system — your labels are the map.
- ☐ Install glass and weatherstrip before final exterior trim.
- ☐ Align doors, hood, and decklid: even gaps, flush fit, correct latch engagement.
- ☐ Torque driveline and suspension fasteners to spec; safety-wire where the factory did.
- ☐ Reconnect wiring by your numbered tags; verify each circuit as you go.
- ☐ Install emblems and trim with correct clips and fasteners — no improvising.
- ☐ Double-check every fluid line routing for chafing, heat exposure, and clearance.
- ☐ Reinstall VIN and trim tags exactly as removed — photograph their original fasteners first.

Shop tip: If a fastener is left over, stop. Find where it goes. Leftover bolts are the car's way of telling you something.

PHASE 11 OF 12

FLUIDS, FIRST START & SHAKEDOWN

The most exciting twenty minutes of the project — earn them with preparation.

- ☐ Fill fluids to spec: high-zinc oil for flat-tappet engines, correct coolant mix, trans and diff fluids.
- ☐ Prime the oil system (drill on the pump drive or pre-luber) until oil reaches the rockers.
- ☐ First start checklist: fire extinguisher handy, watch oil pressure, listen, check for leaks — shut down instantly if anything is wrong.
- ☐ Complete the cam break-in procedure before any idle or driving.
- ☐ Initial timing and carburetor/fuel setup; check for vacuum leaks.
- ☐ Shakedown: 500 miles of varied driving — then full inspection, fluid check, and fastener re-torque.
- ☐ Re-torque cylinder heads per spec after heat cycles, where applicable.
- ☐ Alignment to factory spec after suspension settles.

Shop tip: *Never stand in front of or behind the car on first start. Stand to the side, extinguisher within reach.*

PHASE 12 OF 12

DETAILING & FINAL INSPECTION*Finish like you mean it — this is the car the world sees.*

- ☐ Paint correction: cut, polish, and protect after full cure.
- ☐ Detail the engine bay, undercarriage touch-ups, chrome polish, and glass inside and out.
- ☐ Reapply correct decals, stickers, and inspection marks per your Phase 1 photos.
- ☐ Final nut-and-bolt check: every fastener, every fluid level, every light and switch.
- ☐ Road test: listen for new noises, verify temperatures, brakes, steering, and tracking.
- ☐ Photograph the finished car exactly like Phase 1 — the bookend to the project.
- ☐ File all receipts, the build photos, and this completed checklist with the car's documentation.
- ☐ Keep a punch list during the first month of driving; fix small items before they become permanent habits.

Shop tip: A finished restoration with full documentation is worth more than the same car without it. This binder is part of the car's value.

Project sign-off

When every box in all twelve phases is checked, sign below. Keep this page with the car's documentation.

Vehicle (year / make / model): _____

VIN: _____

Restoration completed: _____

Built by: _____

Date: _____